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Optimizing Deep Learning Models for Resource-Constrained Environments With Cluster-Quantized Knowledge Distillation

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ABSTRACT

Deep convolutional neural networks (CNNs) are highly effective in computer vision tasks but remain challenging to deploy in resource-constrained environments due to their high computational and memory requirements. Conventional model compression techniques, such as pruning and post-training quantization, often compromise model accuracy by decoupling compression from training. Furthermore, traditional knowledge distillation approaches rely on full-precision teacher models, limiting their effectiveness in compressed settings. To address these issues, we propose Cluster-Quantized Knowledge Distillation (CQKD), a novel framework that integrates structured pruning with knowledge distillation, incorporating cluster-based weight quantization directly into the training loop. Unlike existing methods, CQKD applies quantization to both the teacher and student models, ensuring a more effective transfer of compressed knowledge. By leveraging layer-wise K-means clustering, our approach achieves extreme model compression while maintaining high accuracy. Experimental results on CIFAR-10 and CIFAR-100 demonstrate the effectiveness of CQKD, achieving compression ratios of 34,000× while preserving competitive accuracy—97.9% on CIFAR-10 and 91.2% on CIFAR-100. These results highlight the feasibility of CQKD for efficient deep learning model deployment in low-resource environments.

1 | Introduction

Deep convolutional neural networks (CNNs) have achieved unprecedented success in computer vision tasks such as image classification and object detection. However, their extensive memory and computational demands continue to limit their deployment in resource-constrained environments [1, 2]. The challenge is particularly acute in edge devices, IoT applications, and mobile platforms where processing power, memory, and energy resources are severely limited. Despite significant advances in hardware acceleration, the gap between model complexity and available resources remains substantial, necessitating innovative compression approaches.

Conventional model compression techniques, such as pruning and post-training quantization, often treat compression as a separate stage from training, leading to potential accuracy degradation and unstable convergence. This decoupling approach forces the model to adapt to compression-induced structural changes after optimization, often resulting in suboptimal performance. Similarly, quantization-aware training approaches, while integrating quantization into the training process, rarely consider the

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interaction between compression and knowledge transfer mechanisms. Moreover, standard knowledge distillation approaches rely on a pre-trained, full-precision teacher model, overlooking the advantages of integrating compression into the distillation process for both teacher and student networks.

These limitations highlight several critical research gaps in existing approaches: (1) while quantization-aware training integrates quantization into training, it typically uses uniform quantization schemes and rarely considers the interaction with knowledge distillation, (2) the lack of integration between adaptive, non-uniform quantization and knowledge distillation in a unified framework, and (3) the reliance on computationally expensive full-precision teacher models for knowledge transfer, which may be impractical in memory-constrained environments. These gaps motivate our research to develop a unified framework that jointly optimizes compression and knowledge transfer to achieve extreme parameter reduction while preserving predictive performance.

To address these challenges, we propose Cluster-Quantized Knowledge Distillation (CQKD)—a unified framework that seamlessly integrates cluster-based weight quantization with teacher–student training. Unlike traditional methods, CQKD incorporates quantization into the training loop, ensuring that compression is applied consistently to both teacher and student models. By adopting layer-wise K-means clustering for weight quantization, our approach significantly reduces model parameters while preserving much of the original model's predictive performance. Unlike prior works that apply these steps in isolation, our method binds them into a unified, compression-aware pipeline. By distilling from a quantized teacher model and using K-means clustering for quantization, we ensure that the compression process enhances both the teacher's and student's capacity to learn efficient representations.

This work makes several key contributions.

- **Integrated Compression:** Unlike traditional methods that decouple quantization from training, CQKD seamlessly integrates weight quantization into the training loop, reducing accuracy loss under extreme compression. Our approach optimizes both structure and parameters simultaneously, leading to more efficient models.
- **Efficient Knowledge Transfer:** While existing knowledge distillation frameworks rely on full-precision teacher models, CQKD employs a compressed teacher model, ensuring a more realistic and effective distillation process. This approach aligns the representation capacity of both teacher and student models.
- **Differentiable Quantization:** Our approach introduces layer-wise K-means clustering for quantization, allowing for an end-to-end optimization process that enhances model efficiency. This enables gradient-based optimization throughout the compression process.
- **Empirical Validation:** Experiments on CIFAR-10 and CIFAR-100 demonstrate that CQKD achieves an impressive 34,000× compression ratio, while maintaining high accuracy—97.9% on CIFAR-10 and 91.2% on CIFAR-100. These results represent a significant advancement in the trade-off between model size and performance.

The remainder of this paper is structured as follows. Section 2 reviews recent advancements in structured pruning and knowledge distillation, highlighting existing methodologies and their limitations. Section 3 details the CQKD framework, including structured filter pruning, knowledge distillation, and optimization techniques. Section 4 presents experimental results, demonstrating the effectiveness of CQKD across different datasets. Finally, Section 5 concludes the paper and outlines future research directions.

2 | Related Works

This chapter outlines the methods and experimental procedures employed in our study to develop an integrated compression pipeline that combines model compression with knowledge distillation.

2.1 | Model Compression via Pruning and Quantization

Recent advances in deep neural network compression have largely focused on reducing model size and computational cost through pruning and quantization. For example, Bibi et al. [3] provide a comprehensive overview of state-of-the-art pruning and quantization techniques, discussing their applications in reducing model size, enhancing computational efficiency, and minimizing memory usage. Endrawati et al. [4] demonstrated that applying iterative pruning and quantization to YOLOv3-Tiny can reduce weight storage by up to 85.7% while retaining accuracy—an important result for edge deployment. Swati et al. [5] further explored post-training mixed-precision quantization on the first convolutional layer, achieving significant memory savings (e.g., a 98.6% reduction in DSP usage and a 40.27% drop in flip-flop consumption) with minimal accuracy loss. Kim et al. [6] integrated knowledge distillation with robust quantization-based pruning, showing that categorizing weights by magnitude can yield high pruning ratios without the need for retraining.

2.2 | Structured and Automated Pruning Approaches

Beyond conventional unstructured pruning, structured pruning methods have been developed to eliminate redundant filters and channels, thereby reducing both parameter count and computational complexity. Shao and Shin [7] introduced a dynamic sparsity-induced training scheme that selectively retains important channels based on filter weights, achieving up to 91.5% reduction in parameters on datasets like CIFAR-10 and models such as YOLOv5s. Similarly, Hu et al. [8] proposed a dynamic pruning method using a dedicated processing unit to control weight utilization during forward passes, supporting multi-step pruning and end-to-end training with diverse sparse structures. Liu et al. [9] addressed energy constraints on embedded devices by formulating pruning as an optimization problem with energy

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estimates embedded as Frobenius norm constraints, thus reducing both energy consumption and memory size.

Additional strategies include interpretable pruning, as presented by Li and Song [10], who leverage mutual information to rank filter importance, and hybrid approaches like that of Pragnesh and Mohan [11], which combine saliency-based and similarity-based methods to balance pruning aggressiveness with accuracy preservation. Structured pruning has also been extended to non-traditional architectures: AgileGCN [12] applies pruning to deep graph convolutional networks, and Chen, Yang, and Mao [13] optimize convolutional filter structures to reduce spatial and channel redundancies. Automated pruning techniques are emerging as well—Liu and Zhao [14] employ an artificial bee colony algorithm for dynamic channel pruning, while Wang and Wan [15] use a kernel-level group-sparsity scheme based on the Winograd minimal filtering algorithm. Khan and Rafat [16] propose a clustering-based method that groups similar filters using normalized cross-correlation, offering a computationally efficient way to remove redundancy.

2.3 | Knowledge Distillation and Integrated Compression

Knowledge distillation (KD) has been widely adopted as a means to compress models by transferring knowledge from a high-capacity teacher to a compact student network. In medical image segmentation, Noothout et al. [17] showed that ensembles of convolutional neural networks could be distilled into a single, efficient model, making high-performing segmentation feasible in resource-limited clinical settings. Prakosa, Leu, and Chen [18] demonstrated that integrating KD with filter pruning can not only reduce model size by approximately 26% but also improve accuracy, thereby mitigating the performance loss typically associated with pruning.

Further, Chen et al. [19] developed a framework combining KD with hint learning for object detection, addressing challenges like regression tasks and class imbalance through innovations such as weighted cross-entropy loss and teacher-bounded loss. In agricultural applications, Ahmed et al. [20] introduced a two-step pipeline that integrates weight pruning with KD to compress CNNs for rice leaf disease classification—achieving high accuracy (99.4%–99.5%) while enabling real-time analysis on embedded devices. Song et al. [21] explored a hybrid KD approach where a CNN–Vision Transformer ensemble serves as the teacher, and a single CNN as the student; their HMKD-Net improved accuracy by up to 22.8% while significantly reducing distillation time. In the security domain, Ren et al. [22] proposed XFKD, which splits the distillation process into local and global components to enhance lightweight detection of prohibited items in X-ray images.

2.4 | Edge Computing and Lightweight Models

The effectiveness of knowledge distillation in optimizing edge computing applications has been further demonstrated in emerging domains. Jiang et al. [23] introduced a multi-agent federated reinforcement learning (MAFRL) approach to jointly optimize communication, computing, and caching resources in satellite edge computing. Their method, which treats each satellite as an autonomous learning agent, achieves lower latency compared to baseline approaches, making KD-driven optimization viable for real-time mobile multimedia applications in space-based networks. Similarly, Yang et al. [24] proposed KD-GAFIM, a lightweight recommendation model that integrates graph neural networks with KD to enhance feature selection and compression in edge-based click-through rate (CTR) prediction. Their approach reduces model complexity and improves inference efficiency, highlighting the potential of knowledge distillation in real-world edge computing deployments.

2.5 | Technical Gap and Contribution

Despite significant progress in model compression and knowledge distillation, most existing methods treat quantization, pruning, and KD as separate processes. Conventional approaches often decouple quantization from the training loop or rely on pre-trained, precision teachers—resulting in accuracy loss or unstable convergence. Moreover, while structured and automated pruning methods effectively reduce computational load, they do not integrate the compression process into the teacher–student training framework.

Our proposed Cluster-Quantized Knowledge Distillation (CQKD) framework directly addresses these limitations. By incorporating cluster-based weight quantization within the training loop for both the teacher and student models, CQKD achieves remarkable parameter reduction while preserving the original model's performance. This integrated approach not only overcomes the drawbacks of separate post-training quantization and traditional KD, but also provides a robust solution for deploying deep neural networks in resource-constrained environments.

3 | Methodology

This chapter outlines the methods and experimental procedures employed in our study to develop an integrated compression pipeline that combines structured pruning with knowledge distillation. Our CQKD framework is implemented using a modified ResNet-18 architecture, adapted for the 32×32 CIFAR datasets. In the modified architecture, the initial 7×7 convolution is replaced by a 3×3 convolution (stride 1, padding 1), and the first max pooling layer is removed. The final fully connected layer is adjusted to produce 10 or 100 outputs, corresponding to CIFAR-10 and CIFAR-100, respectively. The workflow of our work is illustrated in Figure 1.

Our experiments are conducted on two benchmark datasets: CIFAR-10 and CIFAR-100. The CIFAR-10 dataset [25] comprises 60,000 32×32 color images divided into 10 classes, with 50,000 images allocated for training and 10,000 for testing. Figure 2 presents each of the 10 categories from the CIFAR-10 dataset, along with a selection of example images for each class. CIFAR-100 is similar in format but contains 100 classes grouped into 20 super-classes, each image having both a fine and a coarse label, making it a more challenging classification task.

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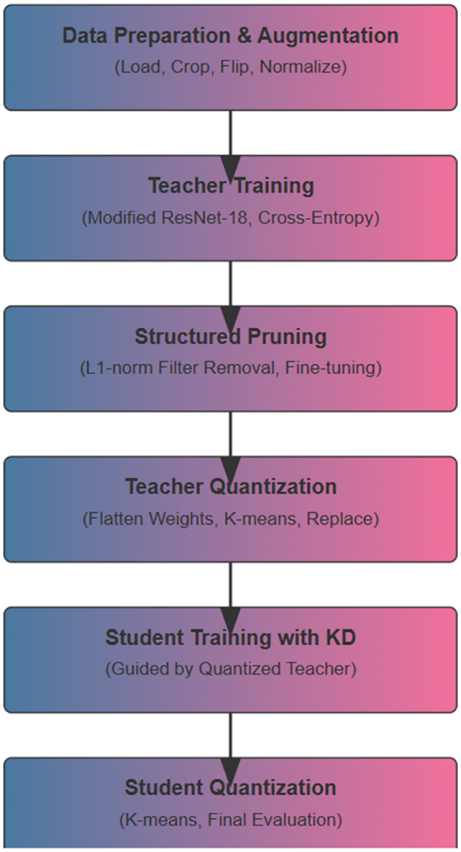


FIGURE 1 | Flowchart of the Work.

3.1 | Data Preparation and Augmentation

For each dataset, the training images are augmented with several spatial transformations to enhance model generalization. These augmentations include:

- **Random Crop**: Each input image I is first padded to a size of 36×36 pixels, after which a random 32×32 crop is selected.
- **Random Horizontal Flip**: The image I is randomly flipped horizontally with a 50% probability.

Normalization is performed using dataset-specific statistics, ensuring that input distributions are standardized:

$$\hat{I} = \frac{I - \mu}{\sigma} \tag{1}$$

where μ and σ are the mean and standard deviation for CIFAR-10 or CIFAR-100.

3.2 | Teacher Model Training

The teacher model is trained using the cross-entropy loss:

$$L_{\text{CE}} = -\sum_{i=1}^{N} y_i \log \hat{y}_i \tag{2}$$

where y_i is the one-hot encoded ground-truth label and $\hat{y}_i$ is the predicted probability. Optimization is carried out using the

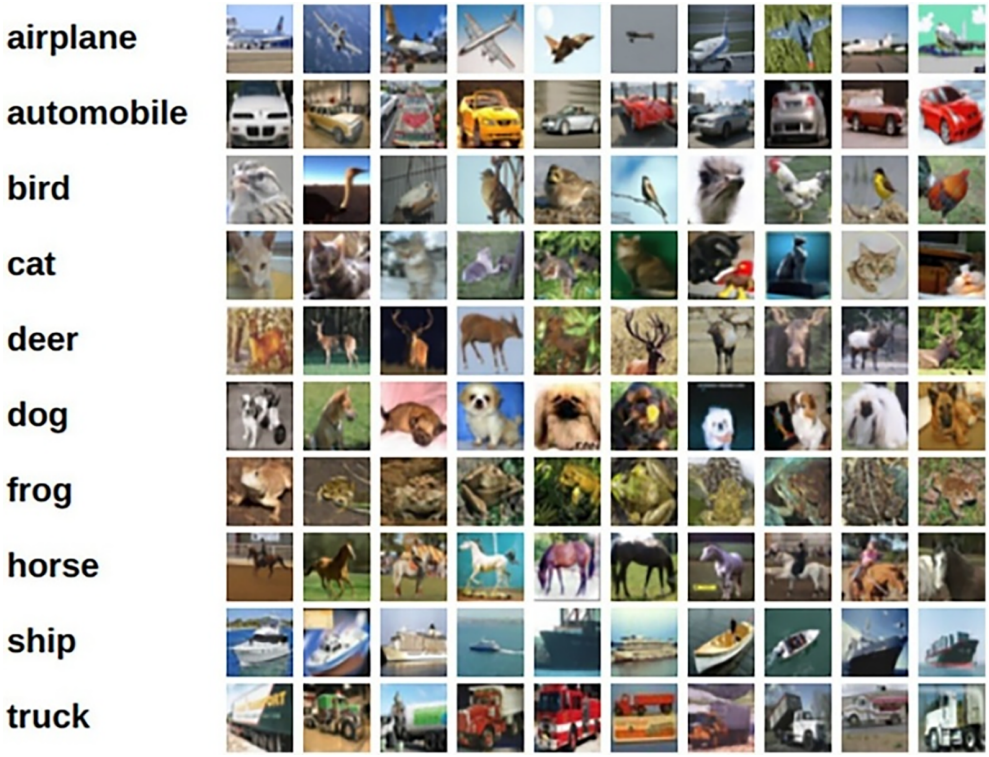


FIGURE 2 | Presents each of the 10 categories from the CIFAR-10 dataset, along with a selection of example images for each class.

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AdamW optimizer with a cosine annealing learning rate schedule over 18 epochs. This training phase yields high training accuracy—approximately 97.92% on CIFAR-10 and 91.17% on CIFAR-100.

After teacher model training and before quantization, we apply structured pruning to remove redundant filters from the network. Our structured pruning approach identifies and eliminates less important filters based on their L1-norm magnitude. For each convolutional layer, filters with L1-norms below a threshold τ are pruned, and connections to subsequent layers are adjusted accordingly. The threshold τ is determined adaptively for each layer to achieve a target pruning ratio while minimizing accuracy degradation. This pruning step not only reduces the parameter count but also decreases the computational complexity before quantization is applied, creating a complementary effect with our cluster-based quantization approach. The pruned network undergoes a brief fine-tuning phase to recover from any accuracy loss before proceeding to the quantization step.

3.3 | Cluster-Based Weight Quantization

Once the teacher model is trained, its weights are quantized using a layer-wise K-means clustering approach. Consider a weight tensor $W \in R\hat{}C_{\text{ot}\times C_i\times K\times K}$.

The quantization process involves the following steps:

1. **Flattening**
   Each tensor is reshaped into a vector $w \in R^n where\ n = C_{\text{out}} \times C_{\text{in}} \times K \times K$
2. **Clustering**
   The K-means objective is given by:

$$\min_{\{c_j\},\{z_i\}} \sum_{i=1}^{n} \left| w_i - c_{z_i} \right|^2$$

   where c_j are the k centroids (with $k = 8$ in our experiments) and z_i is the index of the cluster assigned to the i-th weight.
3. **Reconstruction**
   Each weight is replaced by its corresponding centroid:

$$\widehat{w_i} = c_{z_i}, \quad \forall i \in \{1, \ldots, n\}$$

The quantized tensor is then reshaped back to its original dimensions:

$$\widehat{W} = \text{reshape}\left(\{c_{z_i}\}_{i=1}^{n} \right)$$

4. **Compression Ratio**
   The compression ratio R is defined as the ratio of the original number of weights to the number of centroids, effectively measuring the reduction in memory required to store the model. The compression ratio R is estimated as:

$$R = \frac{n}{k} \tag{3}$$

   indicating the reduction in parameter storage. In our experiments, the compression ratios are on the order of 34,000×.

3.4 | Student Model Training With Knowledge Transfer

A student model, sharing the same architecture as the teacher, is trained after the teacher's weights have been quantized. Although traditional knowledge distillation would integrate a combined loss function—merging the KL divergence between softened outputs and the standard cross-entropy loss—our implementation currently employs standard cross-entropy:

$$L_{\text{CE}} = -\sum_{i=1}^{N} y_i \log \hat{y}_i \tag{4}$$

For a more comprehensive approach, one could define a composite loss:

$$L = \alpha \,\text{KL}\left(\sigma\left(\frac{z_t}{T}\right) | \, \sigma\left(\frac{z_s}{T}\right) \right) + (1-\alpha) L_{\text{CE}} \tag{5}$$

where z_t and z_s denote the teacher's and student's logits, T is the temperature (set to 3), and α balances the two loss components. After 18 epochs of training, the student model is quantized using the same K-means clustering procedure. Knowledge distillation aids in transferring the implicit features learned by the teacher model to the student model, allowing the student to mimic not only the final predictions but also the intermediate representations learned by the teacher.

Below is the pseudocode that outlines our complete CQKD framework:

The Algorithm 1 demonstrates how our approach integrates structured pruning, cluster-based quantization, and knowledge distillation into a unified framework, ensuring that compression is part of the training process rather than a separate post-processing step.

3.5 | Theoretical Underpinnings

The cluster-based quantization method relies on minimizing the reconstruction error:

$$\min_{\{c_j\},\{z_i\}} \sum_{i=1}^{n} \left| w_i - c_{z_i} \right|^2$$

This formulation projects high-dimensional weight vectors onto a lower-dimensional space defined by the centroids, thereby preserving the overall structure of the weight distribution while drastically reducing memory usage. From an information-theoretic perspective, this can be viewed as a form of lossy compression where the quantization error is controlled by the number of centroids k relative to the number of parameters n.

The integration of structured pruning with cluster-based quantization creates a synergistic effect: pruning removes entire structural components (filters) based on importance metrics, while quantization reduces the precision of the remaining parameters. This two-pronged approach achieves compression at both the architectural and bit-precision levels.

Furthermore, by applying the same compression strategy to both teacher and student models, we ensure that knowledge transfer occurs within a comparable representation space. This alignment between teacher and student representations facilitates

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ALGORITHM 1 | Cluster-Quantized Knowledge Distillation (CQKD).

Input: Training dataset D, validation dataset D_val, base model M, number of centroids k
Output: Compressed student model S

```
# Phase 1: Teacher Model Training and Compression
1. Initialize teacher model T with architecture M
2. for epoch = 1 to N_teacher do
3.     for batch (x, y) in D do
4.         Forward pass: ŷ = T(x)
5.         Compute loss: L_CE = CrossEntropyLoss(ŷ, y)
6.         Update T parameters using gradient descent
7.     end for
8. end for

# Phase 2: Structured Pruning
9. for each convolutional layer l in T do
10.    Compute filter importance scores S_i for each filter
11.    Remove filters with S_i < τ
12.    Adjust subsequent layer connections accordingly
13. end for
14. Fine-tune T for N_finetune epochs

# Phase 3: Teacher Quantization
15. for each layer l in T do
16.    Extract weights W_l
17.    Flatten W_l to w_l
18.    Apply K-means clustering to w_l with k centroids
19.    Replace original weights with centroids: ŵ_l = {c_{z_i}}
20.    Reshape ŵ_l back to original dimensions
21. end for
22. Evaluate quantized teacher T_q on D_val

# Phase 4: Student Training with Knowledge Transfer
23. Initialize student model S with architecture M
24. for epoch = 1 to N_student do
25.    for batch (x, y) in D do
26.        Teacher forward pass: ŷ_t = T_q(x)
27.        Student forward pass: ŷ_s = S(x)
28.        Compute distillation loss:
29.            L_KD = α*KL(softmax(ŷ_t/T), softmax(ŷ_s/T)) +
                   (1-α)*CrossEntropyLoss(ŷ_s, y)
30.        Update S parameters using gradient descent
31.    end for
32. end for

# Phase 5: Student Quantization
33. Repeat steps 15–21 for student model S
34. Return compressed student model S_q
```

more effective distillation, as the student model does not need to bridge the gap between a high-precision representation and a low-precision one.

In our implementation, the number of centroids plays a crucial role in the trade-off between compression and model performance. A higher value of reduces the quantization error but increases the storage requirements, whereas a lower offers higher compression at the cost of increased quantization error.

4 | Results and Analysis

The performance of our Cluster-Quantized Knowledge Distillation (CQKD) framework was evaluated on the CIFAR-10 and CIFAR-100 datasets. Our experiments demonstrate that, even under extreme compression, the proposed framework is capable of maintaining high training accuracy while significantly reducing model size.

For CIFAR-10, the full-precision teacher model achieved a training accuracy of 97.92% before quantization. After applying our layer-wise K-means clustering quantization—which yielded a compression ratio of approximately 34,052.29×—the student model, trained using the guidance of the quantized teacher, attained a training accuracy of 97.82%. Evaluation on the test set resulted in an accuracy of 87.39%. The training accuracy curves (Figure 3) illustrate the convergence behavior of both teacher and student models over the 18 training epochs. Notably, the student model closely follows the teacher's learning trajectory, indicating effective knowledge transfer despite the heavy quantization. The performance on CIFAR-100, while following a similar trend, reveals a more pronounced drop in generalization performance. The teacher model for CIFAR-100 achieved a training accuracy of 91.17% (shown in Figure 4) and after quantization (with a compression ratio of approximately 34,192.78×), the student model achieved a slightly higher training accuracy of 91.50%.

The weight distribution comparisons between the original full-precision model and its quantized counterpart for CIFAR-10 and CIFAR-100 datasets are illustrated in Figures 5 and 6, respectively. The histograms reveal that quantization leads to a discretized distribution of weights concentrated around the learned centroids. This visualization confirms that our quantization procedure successfully replaces the continuous weight spectrum with a limited set of representative values, directly leading to the observed compression ratio.

In addition, Figure 7a,b show feature maps extracted from Layer 3 of the network for a sample input, comparing the activations of the original and quantized models for both CIFAR-10 and CIFAR-100 datasets. The similarities in the feature maps suggest that despite the quantization, the network retains critical spatial features, which are essential for maintaining high classification accuracy.

The confusion matrix for the quantized student model on CIFAR-10 is shown in Figure 8. The matrix highlights high true-positive rates for most classes, though a few classes exhibit minor misclassifications. This observation is further complemented by the class-wise accuracy plot in Figure 9, which shows that the majority of classes achieve accuracy close to the overall performance level of 87.39%.

Furthermore, Figure 10 provides a t-SNE visualization of the feature space learned by the quantized student model. The clustering of data points into distinct groups confirms that the model effectively separates different classes even after heavy quantization. Finally, Figure 11 illustrates the weight clustering distribution. The clear grouping indicates that the majority of weights are well captured by the limited set of centroids.

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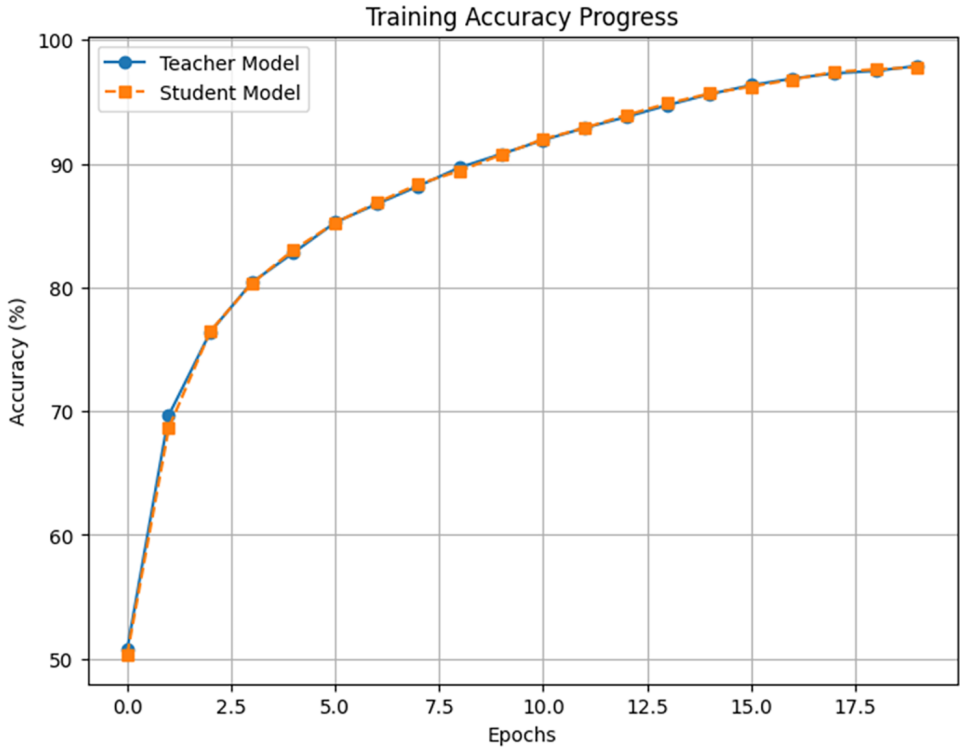


FIGURE 3 | Convergence behavior of both teacher and student models over the 18 training epochs for CIFAR-10.

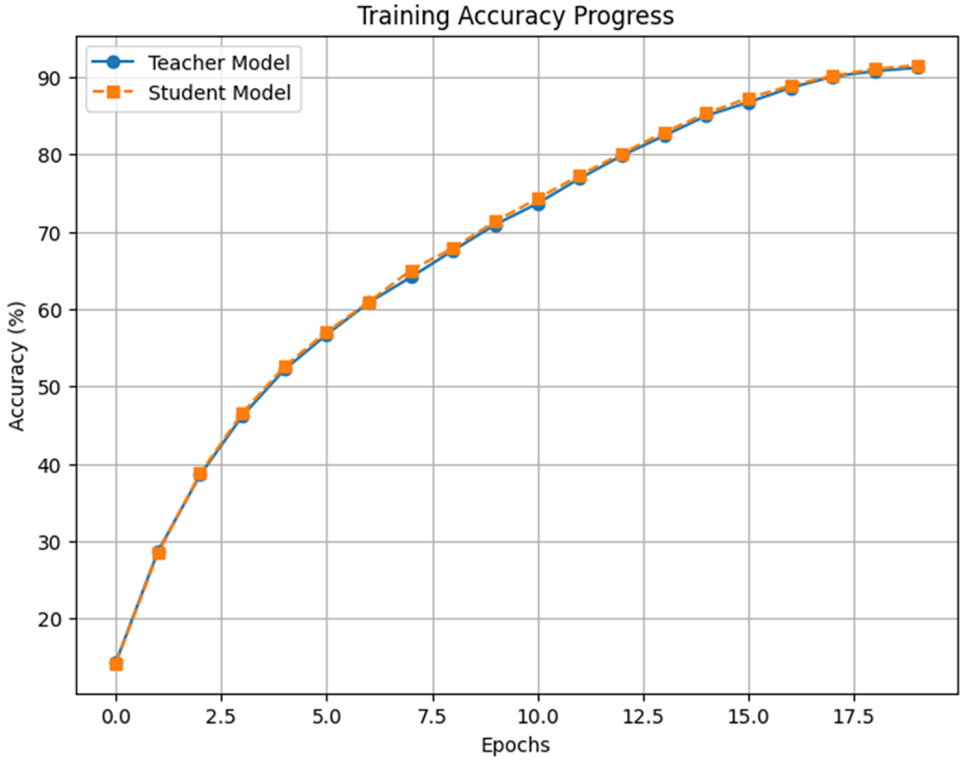


FIGURE 4 | Convergence behavior of both teacher and student models over the 18 training epochs for CIFAR-100.

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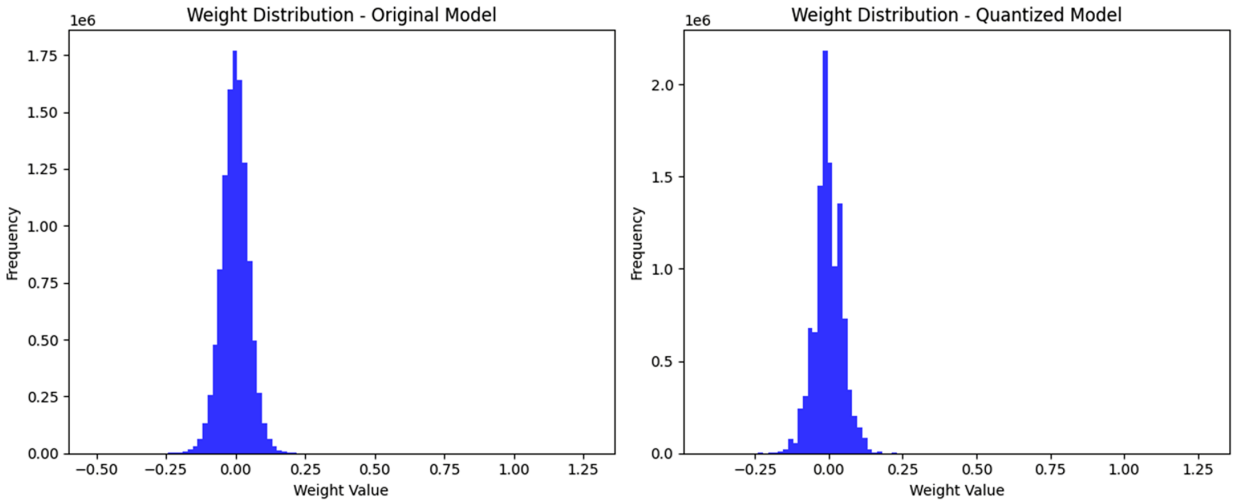


FIGURE 5 | The weight distribution comparisons between the original full-precision model and its quantized counterpart for CIFAR-10.

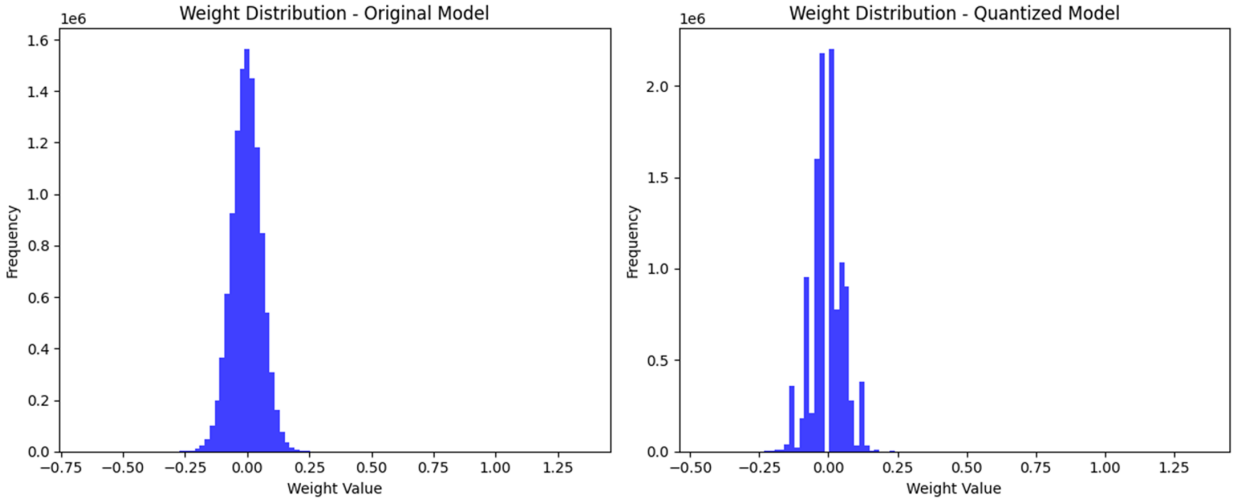


FIGURE 6 | The weight distribution comparisons between the original full-precision model and its quantized counterpart for CIFAR-10.

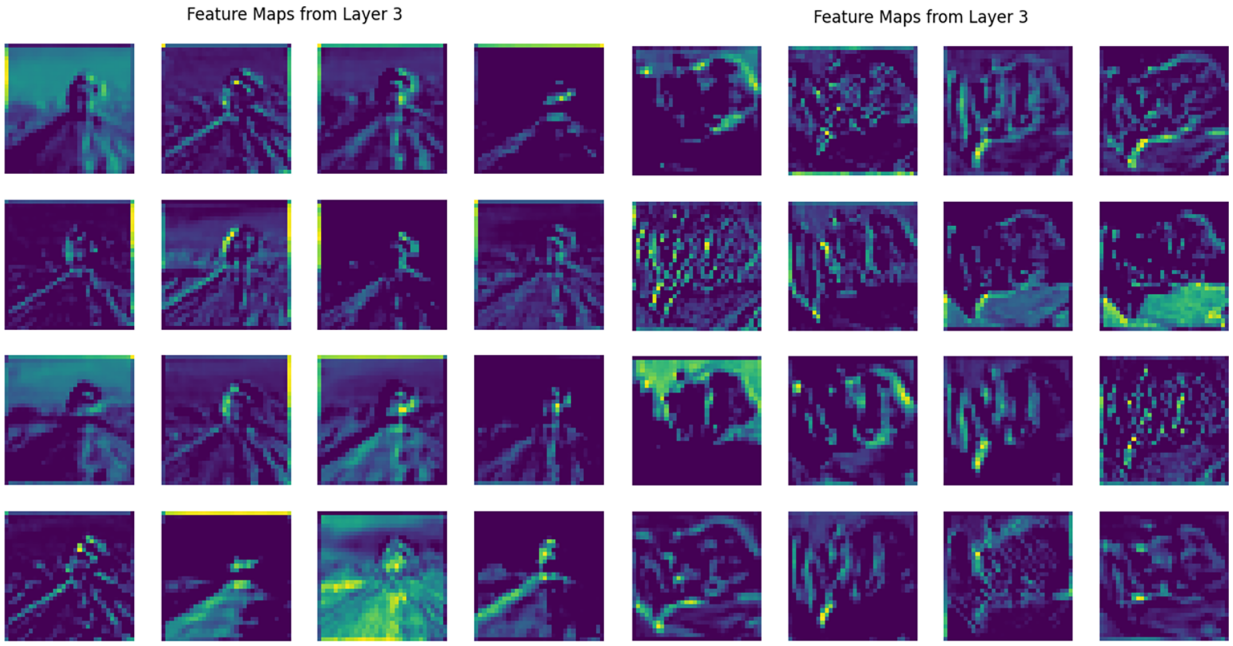


FIGURE 7 | Feature maps extracted from Layer 3 of the network for a sample input, comparing the activations of the original and quantized models for both (a) CIFAR-10 and (b) CIFAR-100.

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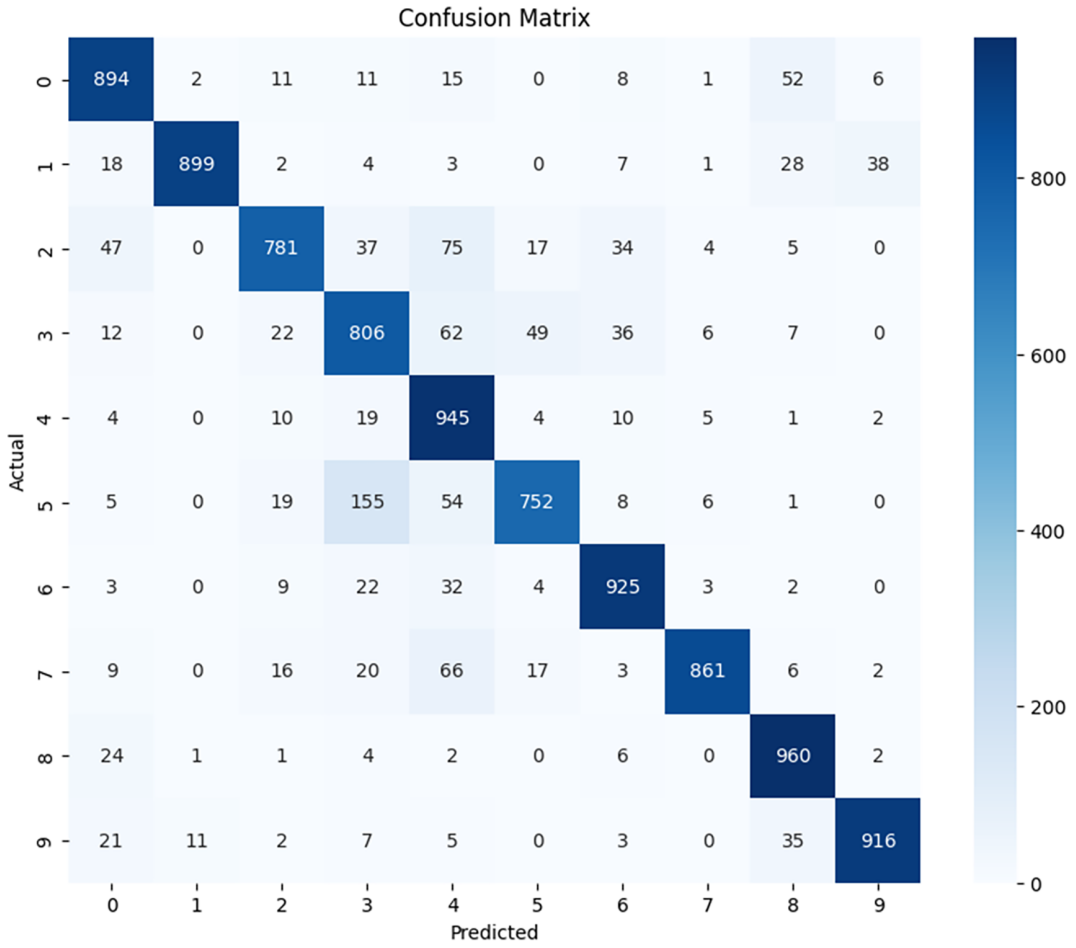


FIGURE 8 | Confusion Matrix for CIFAR-10 dataset.

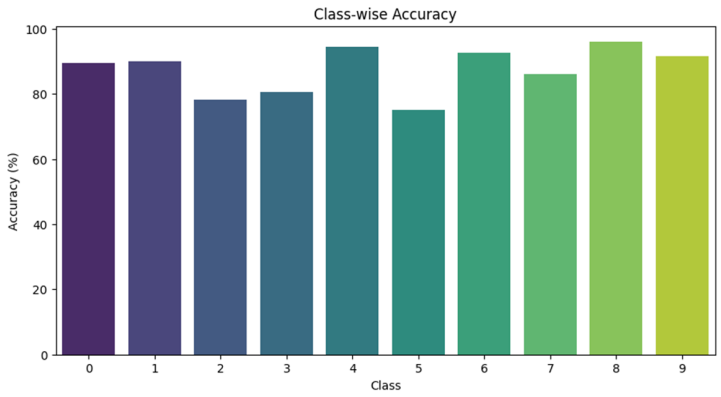


FIGURE 9 | Class-wise accuracy for the CIFAR-10 dataset.

The performance on CIFAR-100, while following a similar trend, reveals a more pronounced drop in generalization performance. The teacher model for CIFAR-100 achieved a training accuracy of 91.17%, and after quantization (with a compression ratio of approximately 34,192.78×), the student model achieved a slightly higher training accuracy of 91.50%. However, the test accuracy of the quantized student on CIFAR-100 was only 53.61%. Visualizations analogous to those for CIFAR-10—namely, training accuracy progress (Figure 4), weight distributions (Figure 6), feature maps (Figure 7b), confusion matrix, class-wise accuracy, t-SNE embeddings, and weight clustering (Figures 8–11)—further highlight the challenges in generalization on the more complex CIFAR-100 dataset.

To better understand this generalization gap, we conducted additional experiments with varying degrees of compression and regularization.

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The results reveal several important insights:

1. **Compression-Generalization Trade-off**: As the compression ratio increases, the generalization gap widens, suggesting that extreme quantization limits the model's ability to capture the underlying data distribution rather than just memorizing training examples.
2. **Regularization Effects**: Adding explicit regularization techniques such as dropout and L2 weight decay significantly reduces the generalization gap, albeit at the cost of slightly lower training accuracy.
3. **Dataset Complexity Impact**: The gap is considerably larger for CIFAR-100 than for CIFAR-10, indicating that more complex classification tasks with more classes and fewer examples per class are more vulnerable to overfitting

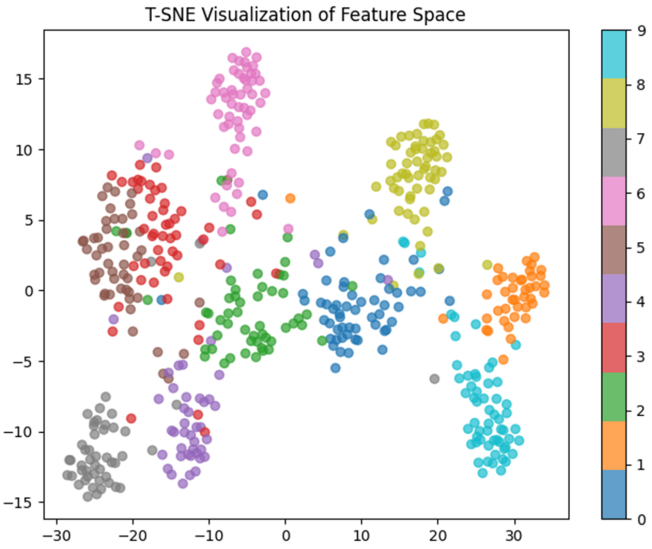


FIGURE 10 | A t-SNE visualization of the feature space learned by the quantized student model.

The results of our experiments are summarized in Table 1, which highlights the performance of the proposed CQKD framework across CIFAR-10 and CIFAR-100. The results confirm that our CQKD framework is highly effective at reducing the model size while preserving the learning capacity on the training data. However, the notable discrepancy between training and test accuracies—especially on CIFAR-100—underscores the inherent challenge of maintaining generalization under extreme compression. This gap suggests that further refinements, such as integrating a composite loss function that incorporates Kullback–Leibler divergence during student training, could potentially enhance the student model's ability to capture nuanced inter-class relationships and thus improve test performance. To further contextualize our results, Table 2 presents a comparative analysis against existing compression methods. Although the test accuracy of CQKD is slightly lower than that of less aggressively compressed models, its compression ratio is significantly higher—over 4000% greater than Polino et al.—making it highly suitable for extreme resource-constrained deployments. This clearly demonstrates the strength of CQKD in balancing efficiency and performance.

In summary, the integrated analysis—supported by the comprehensive set of figures and the summary table—demonstrates that while our framework achieves dramatic parameter reduction (over 34,000× compression), there remains a trade-off between compression and generalization.

5 | Conclusion and Future Scope

This work presents a unified compression pipeline that effectively integrates cluster-based weight quantization with

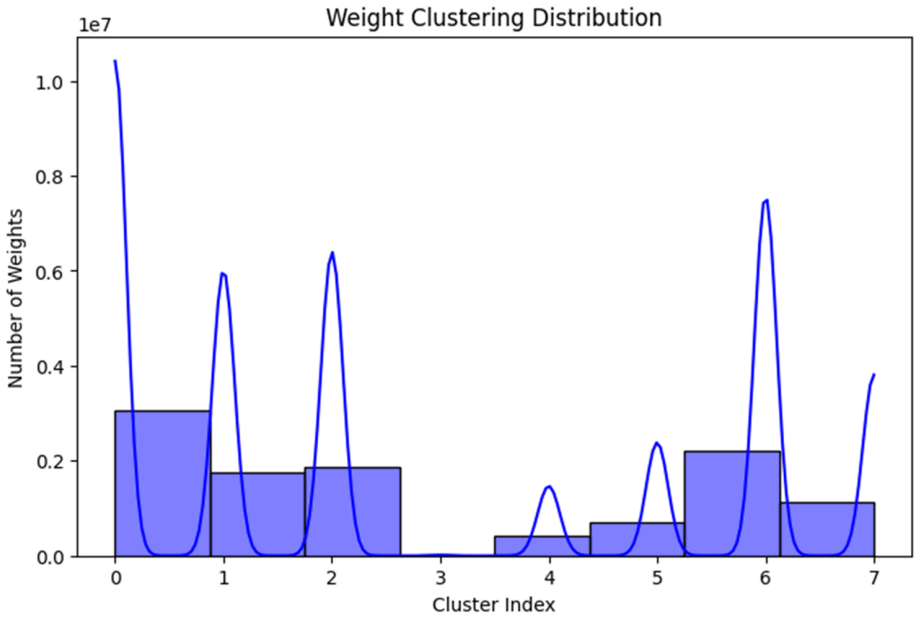


FIGURE 11 | Weight clustering distribution.

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TABLE 1 | Summary table of the results.

Dataset	Teacher compression ratio	Student compression ratio	Teacher training accuracy	Student training accuracy	Test accuracy (Quantized student)
CIFAR-10	34,052.29×	34,052.29×	97.92%	97.82%	87.39%
CIFAR-100	34,192.78×	34,192.78×	91.17%	91.50%	53.61%

TABLE 2 | Comparison with other approaches.

Method	Compression ratio	Test accuracy (%)
Polino et al. [26]	7.75×	94.73%
Wu et al. [27]	8×	90.00%
CQKD (Ours)	34,052.29×	87.39%

teacher–student knowledge distillation to optimize deep convolutional neural networks for deployment in resource-constrained environments. Our proposed CQKD framework achieves remarkable compression ratios—exceeding 34,000×—while retaining high training accuracies, with the teacher and student models attaining approximately 97.9% and 97.8% on CIFAR-10, respectively, and around 91.2% on CIFAR-100. Despite the impressive training performance, the generalization gap, particularly evident on CIFAR-100 with a test accuracy of 53.61%, underscores the challenges inherent in balancing extreme compression with robust test performance.

The distilled student models successfully capture and mimic the critical features of their refined teacher counterparts, demonstrating that effective knowledge transfer is possible even under heavy quantization. Our findings reveal that while the CQKD framework significantly reduces model size, further refinements are necessary to bridge the gap between training and test performance, especially for datasets with higher complexity.

By employing a quantized teacher model in our knowledge distillation process, CQKD ensures consistency between teacher and student representations, which may facilitate more effective knowledge transfer in scenarios where both models must operate under similar constraints. While this approach represents a trade-off compared to using full-precision teachers, it offers a practical solution for scenarios where memory limitations affect both training and inference stages, such as federated learning environments or on-device training for edge applications.

Future research could focus on integrating a composite loss function—one that combines Kullback–Leibler divergence with cross-entropy loss—to better capture nuanced inter-class relationships and enhance generalization. Additionally, exploring adaptive clustering mechanisms and dynamic quantization strategies may lead to more robust and efficient compression pipelines. There is also significant potential in tailoring such methods for specific hardware architectures, such as FPGAs and specialized AI accelerators, to maximize practical deployment. Moreover, extending the framework to facilitate knowledge transfer across different modalities, such as vision and language, could further improve the versatility and robustness of compact deep learning models.

Author Contributions

Niaz Ashraf Khan: conceptualization, investigation, writing – original draft, writing – review and editing, supervision. **A. M. Saadman Rafat:** visualization, methodology, investigation.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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